Development kit for IoT

Synzen Precision Technology has released a new development platform that uses the advanced capabilities of the Nordic nRF9160 module. The development platform features a

new LTE solution, which eliminates the requirement of switching networks, thus reducing costs, complexity, and size of the product. The whole module comes in a small form factor, measuring just 50mm × 50mm. The development platform offers security features to ensure protection of transmission of sensitive data, making it suitable for a wide variety of IoT applications. This new platform is designed to simplify the creation and deployment of IoT projects while ensuring seamless connectivity and secure data transmission.

Synzen Precision Technology

www.synzen.com.tw

Bluetooth SoC

Silicon Labs has released the xG27 family of Bluetooth system on chips (SoC) for small form factor IoT devices. The SoC is based around the ARM Cortex M33 processor and features an

integrated DC-DC boost converter, allowing it to work with voltages as low as 0.8V. It also features an integrated coulomb counter, which enables it to measure the battery state to help users avoid battery depletion during use, thus improving user experience and product safety. The SoC offers IoT device designers energy efficiency, high performance, trusted security, and wireless connectivity. The SoC is ideal for use in tiny, battery-optimised devices like connected medical devices, wearables, asset monitoring tags, smart sensors, and simple consumer electronics like toothbrushes and toys.

Silicon Labs

https://www.silabs.com

Multi-function wireless thermostat

MClimate's has released a connected thermostat that measures movement, humidity, temperature, and illuminance. The standalone thermostat is powered by Epishine's organic indoor solar cell.

The device features a 7.4cm e-ink screen, sensor for movement (PIR), temperature and humidity sensor, LUX sensor, and 3 buttons. The PIR sensor enables the device to detect any movement and control the temperature or humidity of the room. The device can send uplinks after any event or at predetermined intervals. It can be powered by 4 AA batteries or by connecting USB-C, if it needs to be installed in a dark room. The MClimate wireless thermostat can enable energy usage optimisation, thus making buildings smart at a low cost.

MClimate

https://mclimate.eu

SiP for IoT applications

The iMCP HTLRBL32L system-in-package (SiP) from HT Micron can enable communication to an IoT solution. It combines Bluetooth Low

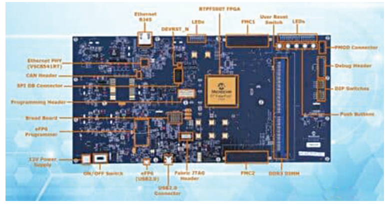
Energy and LoRa to enable easy integration and prototyping of long- and short-range IoT solutions. The product comes in a small package measuring just 13mm × 13mm × 1.1mm. The low-power solution can enable IoT devices to work for years on a single battery charge. It features an accessible MCU and internal memory, and a full range of interfaces for analogue and digital peripherals. It also provides high configurability and flexibility for IoT applications. The SiP can enable designers to build low-cost, low-power, compact IoT solutions that can find application in several fields, including smart farming, smart building, and smart logistics.

HT Micron

www.htmicron.com.br/en

Development kit for space applications

The RT PolarFire development kit is an integrated development platform that combines Microchip's flight-ready RT PolarFire FPGA with the development kit and interfaces. The development platform is suitable for evaluat-



ing design concepts based on actual in-flight electrical and mechanical characteristics. To enable the designers to evaluate high-speed transceivers and test all of their control, DSP, communications, and image-processing algorithms, the kit employs the same package and silicon that final in-orbit or deep-space units will have. The FPGA increases mission-critical computing and connectivity throughput compared to SRAM FPGAs while consuming up to 50% less power and offering greater immunity to radiation-induced configuration single event upsets. This platform will enable designers to solve difficult spaceflight challenges as reducing satellite signal processing congestion. Moreover, it will enable designers to prototype with the same low-power, high-throughput radiation-tolerant (RT) FPGA that will be used in spaceflight.

Microchip Technology

www.microchipdirect.com

T&M EQUIPMENTS

PCIe digitizer cards

Spectrum Instrumentation has added M5i.3350-x16 and M5i.3357-x16 PCIe digitizer cards to the company's flagship M5i series. The cards can turn any PC it into a powerful measurement tool by installing them in a suitable PCIe slot. This allows the users to use the latest CPU and GPU hardware for sig-